

AWS S3 Automation using Python Boto3

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► Situation:

“During my learning of Cloud and DevOps, I wanted to understand how real companies automate cloud tasks instead of manually creating resources from the AWS console every time.

I noticed that creating S3 buckets and uploading files manually takes time and becomes repetitive when working on multiple projects or deployments.”

► Task

“My task was to build a Python automation project that could:

- Connect with AWS services
- Automatically create S3 buckets
- Upload files into the bucket
- Reduce manual cloud operations using automation”

► Action

“I used Python with the Boto3 library, which is AWS SDK for Python.

First, I created an AWS S3 client connection using Boto3.

Then I developed functions for:

- establishing AWS connection,
- creating S3 buckets,
- listing buckets,
- and uploading files automatically.

While building the project, I faced multiple real-world debugging issues such as:

- variable scope errors,
- incorrect AWS parameter names,
- file path issues,
- and function argument mismatches.

I solved them by carefully reading Python tracebacks, understanding AWS API documentation, and improving my debugging skills.”

► Result

“Finally, I successfully built a working AWS S3 automation tool that can create buckets and upload files programmatically.

This project improved my:

- Python scripting skills,
- AWS cloud understanding,
- Boto3 knowledge,
- and real-world debugging ability.

It also helped me understand how DevOps engineers automate cloud infrastructure in

real production environments.”

“I built an AWS S3 automation project using Python and Boto3 to automate bucket creation and file uploads. Through this project, I learned cloud automation, API integration, and real-world debugging of AWS and Python errors.”